

Ch - Thermodynamics

Video #1

- Thermodynamics - Heat, Motion
- About how heat and work are interconverted
- Thermodynamics law: 1, 2, 0
- Earlier heat was considered as a fluid named caloric, which possessed heat properties, and the molecules are self repellant and go away from hot to cold body
- Now: property which transfers from hot to cold body
- Thermodynamics - internal change in system; Macroscopic properties (P, V, T, M, n) due to use of Heat.
[Kinetic Theory of Gases - molecular microscopic level]

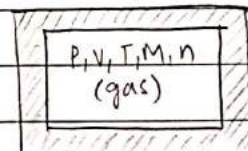
Mechanical Equilibrium -

$$F_{\text{net}} = 0, a_{\text{com}} = 0, \tau_{\text{net}} = 0, \alpha_{\text{com}} = 0$$

Thermal Equilibrium - System macroscopic properties like P, V, T, M, n become constant

- let there be a gas w/ P, V, T, M, n which is insulated from all sides by a non conducting material

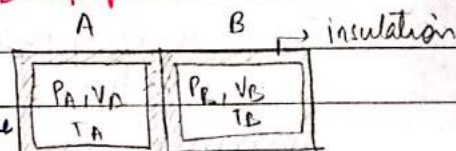
non conducting
→ insulating



- Hence, there will be no exchange of heat, ($\Delta q = 0$) and the system will be adiabatic

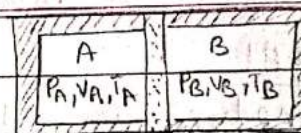
* If the state and properties of a system remain the same; then the system is in Thermal equilibrium [temp of 2 bodies need not be the same!]

2 bodies A, B are separated by non conducting boundary, there will be NO exchange and the system will have same state. $\therefore$ In T_{Eq} even tho $T_A \neq T_B$



☐ Non conducting boundary - adiabatic

☐ Conducting boundary - diathermic

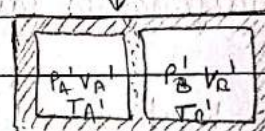


$T_A > T_B$

- Initially $T_A \neq T_B$ and $T_A > T_B$ so KE of particles of A > B.

- Moves faster and transfers heat as KE

- Then $T_A' = T_B'$ and the system now stays in this state of conditions till $t = \infty$ from $t =$ Thermal eq^b



99% Thermal eq^b = $T_A = T_B$

1% " = Macroscopic properties of body stay same

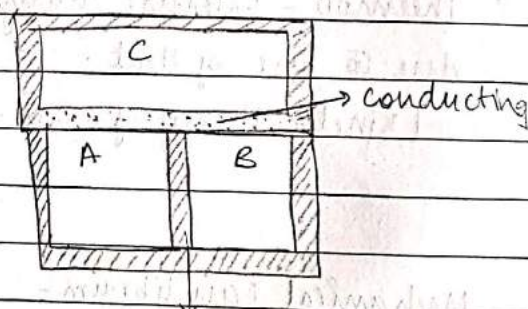
ZEROTH LAW OF THERMODYNAMICS

- At beginning $T_A \neq T_B \neq T_C$
- The boundary b/w A and C (B & C) are conducting so after time t'

$$T_A' = T_C', \quad T_B' = T_C'$$

- when the boundary b/w A & B is made conducting at that instant, it is seen that there is no exchange b/w A' and B'

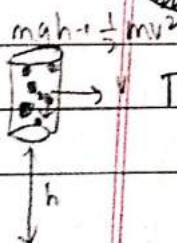
$$\therefore T_A' = T_B'$$



- $\therefore$ If 2 bodies are in thermal eq^b w/ the same body, then they are at TE w/ each other.

Video #2

INTERNAL ENERGY



The total energy possessed by a body = KE + PE + Internal Energy

motion of body

Molecular level Energies

Gravitational + Electrostatic, etc PE of body

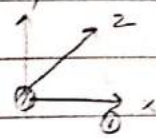
→ Internal Energy accounts for molecular level of energies :-

* KE of molecules (due to v in different axes)

* PE of molecules (due to force of attractions)

* Vibrational / Rotational / Translational KE

considered significant only @ High T



Our considerations are for a body ^{-for ideal gas-} whose $KE = PE = 0$ [at rest and not @ height]

So, $\boxed{\text{Total Energy of body} = \text{Internal Energy}}$
(E) (U)

For ideal gas :- No force of attraction b/w particles, so no PE created due to it

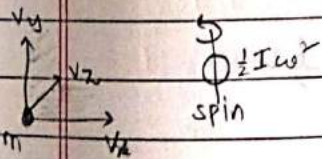
$\therefore \text{TE of ideal gas} = \text{KE of gas} = \text{IE}$
linear rotatory vibration

Idea 1 gas

Monoatomic

eg He, Ne, Ar, Xe, Kr, Rn

$\boxed{E = U = KE}$



$$U = \frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}mv_z^2 + \frac{1}{2}I\omega^2 \rightarrow 0 \text{ (as } I \approx 0)$$

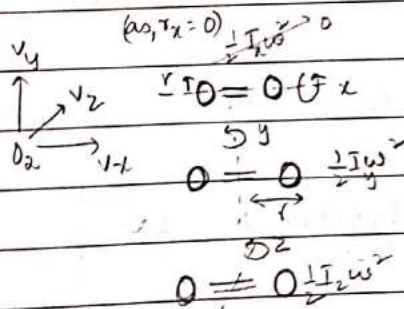
• Vibration later

$\boxed{DOF = 3}$

(3 independent terms)

Diatomic

eg O_2, N_2, H_2 , etc



$$\therefore U = \frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}mv_z^2 + \frac{1}{2}I_x\omega_x^2 + \frac{1}{2}I_y\omega_y^2 + \frac{1}{2}I_z\omega_z^2$$

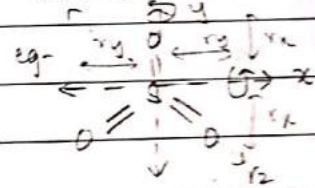
$\boxed{DOF = 5}$

Polyatomic

eg SO_2, SO_3, CH_4, NH_3 , etc

Translational = 3

Rotation = 3



$$U = \frac{1}{2}m(v_x^2 + v_y^2 + v_z^2) + \frac{1}{2}I(\omega_x^2 + \omega_y^2 + \omega_z^2)$$

$\boxed{DOF = 6}$

For linear stuff eg-
 $Cl-Be-Cl, I_x \approx 0$ ($r_x = 0$)
 $\therefore DOF = 5$

Degree of freedom - No of independent terms in the expression of KE (doesn't depend on each other at all)

At high temp vibration terms are also added (+2) for each gas

LAW OF EQUIPARTITION OF ENERGY

lay man lang: The total KE / IE of the body is divided equally among the each element of degree of freedom

eg: IE = 90J for monoatomic gas: $\frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}mv_z^2 = 30J$

Def. The avg KE associated w/ each degree of freedom is $\frac{1}{2}kT$ where k is Boltzmann's constant, T = absolute Temp.

$$k \text{ (Boltzmann constant)} = \frac{R \text{ (gas constant)}}{N_A \text{ (Avogadro NO } N_A)}$$

$$k = \frac{8.314}{6.023 \times 10^{23}}$$

Average KE associated w/ each DOF of a monoatomic gas = $\frac{1}{2}kT$
(UE) for all the gas = $\frac{3}{2}kT$

(for mono, $f=3$)

$$U \text{ monoatomic mole} = \frac{f}{2}kT = \text{KE of mole}$$

$$KE_{\text{mol}} = U_{\text{mole}} = \frac{f}{2}kT \times N = \frac{f}{2}RT \quad ; \quad U_{\text{moles}} = \frac{f}{2}RT \times n$$

$$\therefore \text{Internal Energy 'U' } \frac{f}{2}RT$$

Summary

- $E_{\text{molecule for 1 DOF}} = \frac{1}{2}kT$
- $E_{\text{1 molecule}} = \frac{f}{2}kT$
- $E_{\text{1 mole}} = \frac{f}{2}kT \times N_A = \frac{f}{2}RT$
- $E_{\text{n mole}} / U_{\text{n mole}} = n \times \frac{f}{2}RT$

Q) Find KE of 2 moles of H_2 at 300K
diatomic gas, $KE = U = \frac{f}{2} KT$ for 1 mol^o

$$KE = \frac{f}{2} KT \times N_A \times 2 \text{ for 'n' moles} = n \frac{f}{2} RT = 2 \times \frac{5}{2} \times RT = 5RT$$

Q) Find KE of 1 mol^o of He at 300K

$$KE = U = f \times \frac{1}{2} KT = 3 \times \frac{1}{2} KT = \frac{3}{2} KT$$

$$\star U = KE = n \cdot \frac{f}{2} RT \quad (1 \text{ mole})$$

[Internal Energy is state fun]
depends on state
NOT Path

diff.iating both sides

$$dU = n \left(\frac{f}{2} R \right) dT \quad (\text{as 'n', 'f' = constant can't diff})$$

$$\Delta U = n \left(\frac{f}{2} R \right) \Delta T$$

$$\Delta U = n C_V \Delta T$$

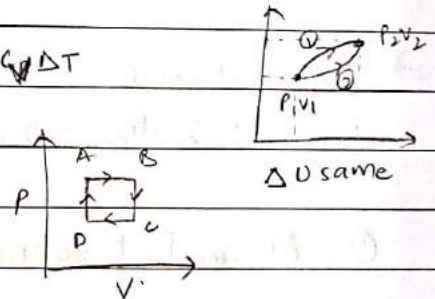
$$\star \Delta U = U_f - U_i \quad (\text{doesn't depend on path})$$

$$\therefore \text{cyclic process } \Delta U = 0$$

$$\star \text{for constant temperature } \Delta T = 0$$

$$\text{or, } \Delta U = 0$$

[Internal Energy is function of T]



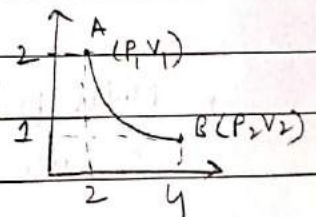
Q) Find ΔU in the case where a gas changes from state A to B

$$P_1 V_1 = 2 \times 2 = 4$$

$$P_2 V_2 = 1 \times 4 = 4$$

as $P_1 V_1 = P_2 V_2$, Boyle's law is obeyed

$$\therefore \Delta T = 0 \quad \text{or } \Delta U = 0$$



Video #3 - Specific Heat Capacity

Heat $\rightarrow$ Energy exchange b/w system and surroundings

$\rightarrow$ Form of transfer of energy (when there is temp diff)

$\rightarrow$ Symbol (Q), unit $\rightarrow$ Joule, common unit: calorie

$$1 \text{ cal} = 4.182 \text{ J}$$

- Heat given to the system (+ve)
Heat given by the system (-ve)

Specific Heat capacities of gases

$$C = \frac{Q}{m\Delta T}$$

Temp needed to raise temp of unit mass by unit °temp

(i) Isothermal, $T = \text{constant}$

$$\Delta T = 0$$

Process

(ii) Adiabatic process, $\Delta q = 0$

$$C_{\text{isothermal}} = \infty$$

$$C_{\text{adiabatic}} = 0$$

① At constant volume

$$C_v = \frac{\Delta}{m\Delta T}$$

1g gas → Temp 1°C ↑ (@ const vol^m)

$$C_v = \text{J/g}^\circ\text{C}$$

② At constant pressure

$$C_p = \frac{\Delta}{m\Delta T}$$

$$C_p = \text{J/g}^\circ\text{C}$$

→ Molar specific Heat capacity - MSHC of gas = Amount of heat needed to raise the temperature of 1 mole of substance by 1°C

→ Relation b/w ~~C_p , C_v~~ C_p , c_p (Molar specific Heat Capacity, ^{gram} specific heat capacity)

$$C_p = \text{Molar mass } c_p$$

$$C_v = \text{Molar mass } c_v$$

For any gas

$$C_p - C_v = R \quad \text{"capital"}$$

g) For N_2 the value of $C_p - C_v =$

(a) $14R$ (b) $\frac{R}{14}$ (c) $28R$ (d) $\frac{R}{28}$

$$C_p - C_v = R$$

$$\text{Molar mass } (C_p - C_v) = R$$

$$C_p - C_v = \frac{R}{28}$$

$$Q_v = nC_v \Delta T$$

$$Q_p = nC_p \Delta T$$

$$C_v = \frac{f}{2} R$$

$$\therefore C_p = \left(\frac{f}{2} + 1\right) R$$

$$\gamma = \frac{C_p}{C_v}$$

$f = 3, 5, 6$ (for mono di, tri atomic)

f

$$C_v = \frac{f}{2} R$$

$$C_p = C_v + R$$

$$\gamma = \frac{C_p}{C_v}$$

$$C_p - C_v = R$$

Monoatomic
(3)

$$C_v = \frac{3}{2} R$$

$$C_p = \frac{5}{2} R$$

$$\gamma = \frac{5}{3} = 1.66$$

Diatomic
(5)

$$C_v = \frac{5}{2} R$$

$$C_p = \frac{7}{2} R$$

$$\gamma = \frac{7}{5} = 1.4$$

Triatomic/Poly
(6)

$$C_v = 6R$$

$$C_p = 4R$$

$$\gamma = \frac{4}{3} = 1.33$$

$$\gamma_{\text{mono}} > \gamma_{\text{di}} > \gamma_{\text{poly}}$$

$$1.66$$

$$1.4$$

$$1.33$$

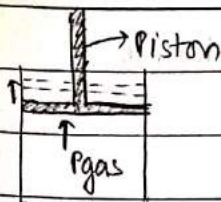
$$\gamma = 1 + \frac{2}{f}$$

Video #4 - Work done in thermodynamics

Work - Mechanical way of transfer of energy.

$$\text{Work done} = P \Delta V$$

pressure $\quad \quad \quad$ change in vol^m

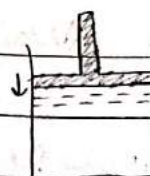


i) If gas expands = work done by gas

$$\Delta V = V_f - V_i = +ve$$

$$W = +ve$$

in chemistry reverse signs



ii) If gas compresses, work done on gas (system)

$$W = P \Delta V \quad (V_i > V_f)$$

$$= -ve$$

gas Expansion is having +ve work!

Q) Find the work done by a gas ($P = 1 \text{ atm}$), when it expands 0.5 L

$$\begin{aligned} w &= P \Delta V = 1 \text{ atm} (0.5 \text{ L}) \\ &= 0.5 \text{ atm-L} \\ &= 0.5 \times 101.3 \text{ J} \\ &= 50.65 \text{ J} \end{aligned}$$

Q) Find the work done?

$$\begin{aligned} w &= P \Delta V \\ &= ? \end{aligned}$$

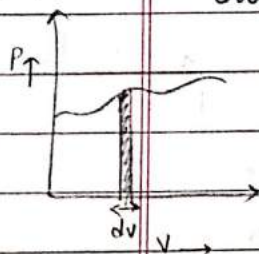
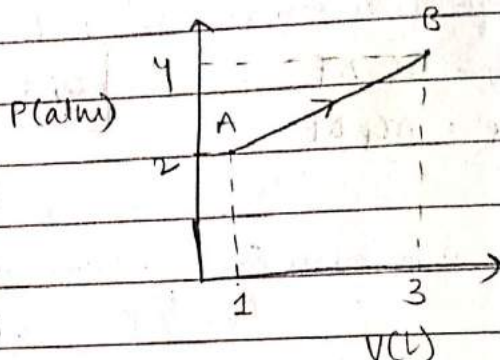
but pressure is not constant

$$w = P \Delta V \text{ (in strip - } P \text{ constant)}$$

$$dw = P dv$$

$$\int dw = \int P dv \Rightarrow \boxed{w = \int P dv}$$

Here work = area under graph = $\oplus 6 \text{ atm-L}$ (gas expand)



Q) Find the work done b/w ABC

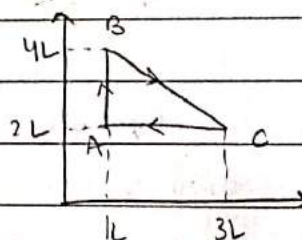
→ work done = path function

$$\therefore w = \text{Area under } AB + BC + AC$$

$$w_{AB} = 0 \text{ (Vol}^m \text{ same } \Delta V = 0)$$

$$w_{BC} = \text{area of trapezium} = \frac{1}{2}(2)(4+2) = \oplus 6 \text{ atm-L}$$

$$w_{CA} = \text{area of rectangle} = \frac{1}{2}(2)(2) = \ominus 2 \text{ atm-L (Vol}^m \text{ less / gas contracted)}$$



$$\text{Total} = 4 \text{ atm-L} = 4 \times 101.3 \text{ J} = 405.2 \text{ J}$$

→ Here, $\Delta V = 0$ (cyclic)

Shortcut in physics for 'w' done

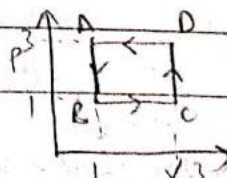
$$\boxed{w = + \text{Area of cycle}}$$

→ +ve if clockwise

→ -ve if anticlockwise

Q) $w = ?$ anticlockwise

$$\therefore w = -(\text{area}) = (-4 \text{ atm-L})$$



Video #5 - First Law of Thermodynamics (Law of Conservation of Energy)

$$\Delta U = \Delta q + \Delta w$$

change in Internal Energy Heat given to the system work done by the system

$$dE/du = dq + dw$$

INTERNAL ENERGY

Total energy = Internal Energy (when KE/PE of body = 0)

$$Q = w + \Delta U$$

Heat given to the system Work done by system Increase in Internal Energy of system

→ Heat given used to increase 'U' or do work

→ Q
 given to system (+ve)
 given by system (-ve)

$$Q = nC\Delta T$$

$Q_V = nC_V\Delta T$ (with $\frac{f}{2}R$)
 $Q_P = nC_P\Delta T$ (with $(\frac{f}{2}+1)R$)

$$R = 8.314 \text{ J/mol K}, 2 \text{ cal/mol K}$$

→ w
 by system = (+ve)
 on system = (-ve)

$$\Delta U = n\left(\frac{f}{2}kT\right)N_A = n\frac{f}{2}R\Delta T$$

$\xrightarrow{C_V}$

$$\Delta U = nC_V\Delta T$$

→ state function, work in cyclic = 0, path independent

Q) 1 mole of O_2 is heated from $25^\circ\text{C} \rightarrow 125^\circ\text{C}$ at constant volume. Find (i) Q (ii) ΔU (iii) Δw [Given, O_2 is diatomic, $f=5$, $R=2 \text{ cal/mol}$]

$$\Delta T = 100 \text{ K}$$

$$\Delta U = nC_V\Delta T = 1 \times \left(\frac{f}{2}R\right)\Delta T = 1 \times \frac{5}{2} \times 2 \times 100 = 500 \text{ cal}$$

$$\Delta w = 0 \quad (\Delta U=0)$$

$$\Delta Q = 500 \text{ cal} = \Delta U$$

Q) Above question w) constant pressure and 1 mole He

$$U = nC_V \Delta T$$

$$= n \frac{f}{2} R \Delta T$$

$$= 1 \times \frac{3}{2} \times 2 \times 100 = 300 \text{ cal}$$

$$\therefore W = 200 \text{ cal}$$

$$Q = Q_p = nC_p \Delta T$$

$$= n \left(\frac{f}{2} + 1 \right) R \Delta T$$

$$= 1 \times \left[\frac{3}{2} + 1 \right] 2 \times 100$$

$$= \frac{5}{2} \times 2 \times 100 = 500 \text{ cal}$$

Q) In an isothermal process, 50J of heat is given to an ideal gas. Find w & ΔU

Ans: T-constant ; $Q = w + \Delta U = w + nC_V \Delta T \rightarrow 0$
 $Q = 50 \text{ J} = w$

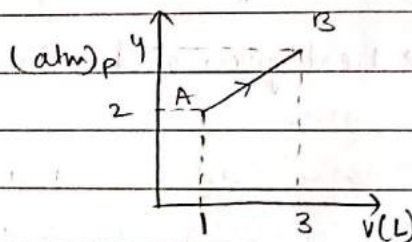
Or all the energy provided in isothermal process used to do work

Q) Find $w_{AB} = ?$ if $\Delta U = +200 \text{ J}$, $Q = ?$

$$w_{AB} = \text{area} = 6 \text{ atm-L} = 606 \text{ J}$$

$$Q = w + \Delta U$$

$$= 606 + 200 = 806 \text{ J}$$



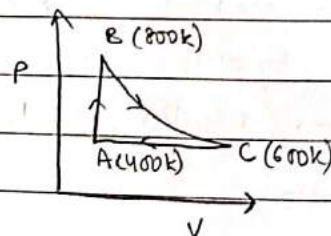
JEE MAINS Q) 1 mole diatomic gas has following path then -
2014

(a) $\Delta U_{ABCA} = 250 \text{ R}$

(b) $\Delta U_{CA} = 750 \text{ R}$

(c) $\Delta U_{AB} = -350 \text{ R}$

(d) $\Delta U_{BC} = -500 \text{ R}$



Soln.

$$w_{ABCA} = \text{area} = \oplus, \text{ but } \Delta U = 0 \text{ (cycle)}$$

$$\Delta U_{AB} = nC_V \Delta T = n \frac{f}{2} R \Delta T = 1 \times \frac{5}{2} \times 2 \times 400 = 4000 \text{ J} = +1000 \text{ R}$$

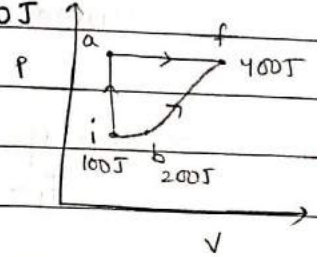
$$\Delta U_{BC} = n \frac{f}{2} R \Delta T = 5 \times 200 = -1000 \text{ J} = -500 \text{ R}$$

$$\Delta U_{CA} = -200 \times 5 = -500 \text{ R}$$

52:41

JEE Advance Q) Find $\frac{Q_{bf}}{Q_{ib}}$ if $U_i = 100\text{J}$, $U_b = 200\text{J}$, $Q_{iaf} = 500\text{J}$,

$$W_{af} = +200\text{J}, W_{ib} = +50\text{J}, W_{bf} = +100\text{J}$$



(a) 1 (b) 2 (c) 3 (d) 4

$$U_{ib} = U_b - U_i = +100\text{J}, W_{ib} = 50\text{J}$$

$$\therefore Q_{ib} = 150\text{J}$$

$$Q_{iaf} = W_{iaf} + U_{if}$$

$$500 = W_{ia} + W_{af} + U_{if} = 200 + U_{if}$$

$$300 = U_{if} = U_{iaf}$$

$$300 = U_f - U_i \Rightarrow U_f = U_i + 300 = 400\text{J}$$

ib	bf
$\Delta U = 100\text{J}$	$\Delta U = 200\text{J}$
$W_{ib} = 50\text{J}$	$W = 100\text{J}$
$Q = 150\text{J}$	$Q = 300\text{J}$
Ratio = $\frac{300}{150} = 2$	

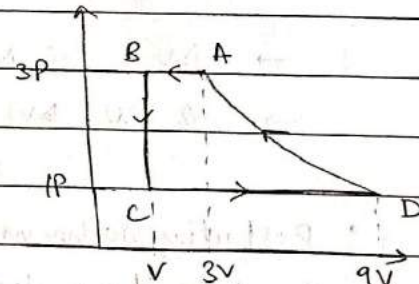
JEE 2011 Q)

Process

options

- a) $A \rightarrow B$ (T, q, x) p) U increases
 b) $B \rightarrow C$ (q, r) q) U decreases
 c) $C \rightarrow D$ (p, e) r) Q is lost
 d) $D \rightarrow A$ (t, y) s) Q is gained

t) work is done on system



→ t) work is done on system (t) → a) expand - by

$$A \rightarrow B \quad 3P \times 2V = 6PV \quad (\text{work} = -ve)$$

No. of moles in all points is equal

$$A: 3P(3V) = nRT \Rightarrow \frac{9PV}{RT} = n$$

$$B: 3P(V) = nRT \Rightarrow \frac{3PV}{RT} = n$$

$$W = \text{compress} (-ve) \quad Q = nC_p \Delta T \quad U = nC_v \Delta T$$

$$PV \downarrow \therefore T \downarrow \quad (PV = kT)$$

$$\text{or, } U \text{ - decrease} \\ Q = U + W = -ve$$

• $B \rightarrow C$ (volume same $\therefore W = 0$)

$$P_1 V_1 = 3PV \quad P_2 V_2 = PV$$

$$\therefore P_1 V_1 \downarrow \text{ or, } T \downarrow \therefore U \text{ decreases, } \therefore Q = \text{decrease}$$

• $C \rightarrow D$ (volume $\uparrow \therefore W = +ve$)

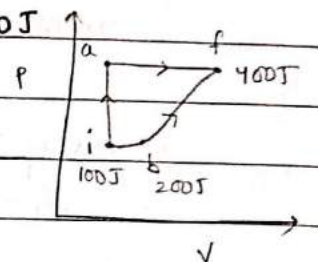
$$P_1 V_1 = PV \quad P_2 V_2 = 9PV \quad \text{or, } PV \uparrow \therefore T \uparrow \quad [U \uparrow, Q = U + W = +ve]$$

• $D \rightarrow A$ (volume $\downarrow$, $W = -ve$) , $P_1 V_1 = P_2 V_2 = 9PV$ (isothermal) $U \rightarrow \text{constant}$
 $Q = W + \Delta U = -ve$ (lost)

JEE Advance Q) Find Q_{bf} if $U_i = 100\text{J}$, $U_b = 200\text{J}$, $Q_{iaf} = 500\text{J}$,

Q_{ib}

$$W_{af} = +200\text{J}, W_{ib} = +50\text{J}, W_{bf} = +100\text{J}$$



(a) 1 (b) 2 (c) 3 (d) 4

$$U_{ib} = U_b - U_i = +100\text{J}, W_{ib} = 50\text{J}$$

$$\therefore Q_{ib} = 150\text{J}$$

$$Q_{iaf} = W_{iaf} + U_{if}$$

$$500 = \cancel{W_{ia}} + W_{af} + U_{if} = 200 + U_{if}$$

$$300 = U_{if} = U_{iaf}$$

$$300 = U_f - U_i \Rightarrow U_f = U_i + 300 = 400\text{J}$$

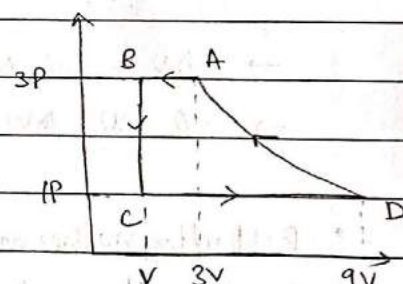
ib	bf
$\Delta U = 100\text{J}$	$\Delta U = 200\text{J}$
$W_{ib} = 50\text{J}$	$W = 100\text{J}$
$Q = 150\text{J}$	$Q = 300\text{J}$ (given)
Ratio = $\frac{300}{150} = 2$	

JEE 2011 Q)

Process

Options

- a) $A \rightarrow B$ (t, q, x) p) U increases
 b) $B \rightarrow C$ (q, r) q) U decreases
 c) $C \rightarrow D$ (p, e) r) Q is lost
 d) $D \rightarrow A$ (t, x) s) Q is gained
 t) work is done on system



→ t) work is done on system ΔV (tent) → a) expand by d) $A \rightarrow B$ (work = -ve) $3P \times 2V = 6PV$

No. of moles in all points is equal

$$\begin{aligned} \bullet \quad A: 3P(3V) &= nRT \Rightarrow \frac{9PV}{RT} = n \\ B: 3P(V) &= nRT \Rightarrow \frac{3PV}{RT} = n \end{aligned} \quad \left. \begin{array}{l} W = \text{compress (-ve)} \\ Q = nC_p \Delta T \\ U = nC_v \Delta T \end{array} \right\} \begin{array}{l} PV \downarrow \therefore T \downarrow (PV = kT) \\ \text{or, } U \text{ decreases} \\ Q = U + W = -ve \end{array}$$

• $B \rightarrow C$ (volume same $\therefore W = 0$)

$$P_1 V_1 = 3PV \quad P_2 V_2 = PV$$

$$\therefore P_1 V_1 \downarrow \text{ or, } T \downarrow \therefore U \text{ decreases, } \therefore Q = \text{decrease}$$

• $C \rightarrow D$ (volume $\uparrow \therefore W = +ve$)

$$P_1 V_1 = PV \quad P_2 V_2 = 9PV \quad \text{or, } PV \uparrow \therefore T \uparrow [U \uparrow, Q = U + W = +ve]$$

• $D \rightarrow A$ (volume $\downarrow$, $W = -ve$), $P_1 V_1 = P_2 V_2 = 9PV$ (Isothermal) $U \rightarrow \text{constant}$
 $Q = W + \Delta U = -ve$ (lost)

Video #6

Summary:

Adiabatic process ($\Delta Q = 0$)	Isothermal ($\Delta T = 0$)
→ gas of 'n' moles $T_1 \rightarrow T_2$	$W = nRT \log_e \frac{V_2}{V_1}$
$W = \frac{nR(T_2 - T_1)}{\gamma - 1}$	

TYPES OF PROCESSES

① Isothermal Process -

- Constant Temperature, $\Delta T = 0$

ideal gas: $P \propto \frac{1}{V} \Rightarrow PV = \text{constant} [P_1 V_1 = P_2 V_2]$

→ $\Delta U = nC_v \Delta T \quad \therefore \text{Isothermal process } \Delta U = 0$

→ $Q = \Delta U + W \quad \therefore \text{Isothermal process } Q = W$

(Heat used to do work, not increase temperature)

- Perfectly isothermal process - very slow
- The boundary should be perfectly conducting / diathermic
- Slope of P-V graph

$PV = \text{constant}$

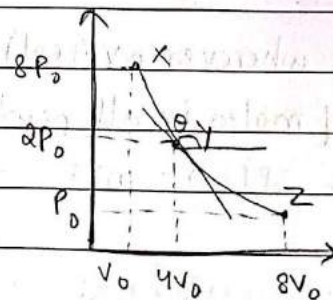
$d(PV) = 0$

$V dP + P dV = 0$

$P dV = -V dP$

$\frac{-P}{V} = \frac{dP}{dV}$

$\therefore \text{Slope of P/V graph in isot. process } \left(\frac{dP}{dV} \right)_{\text{iso}} = -\frac{P}{V}$ -ve sign as slope -ve



eg- Find slope at Y: $\left(\frac{dP}{dV} \right)_Y = -\frac{P}{V} = -\frac{4P_0}{4V_0} = -\frac{P_0}{V_0}$

work done in Isothermal process (by the gas)

$$W = P \Delta V \quad \text{if } P = \text{constant}$$

$$W = \int P dV \quad \text{if } P = \text{not constant}$$

At constant temp an ideal gas expands from V_1 to V_2 and $P_1 \rightarrow P_2$

$$W = \int_{V_1}^{V_2} \frac{nRT}{V} dV$$

$$\Rightarrow W = nRT \int_{V_1}^{V_2} \frac{1}{V} dV = nRT \log_e \frac{V_2}{V_1} = 2.303 \times nRT \log_{10} \frac{V_2}{V_1}$$

Formulae:-

$$W = nRT \log_e \frac{V_2}{V_1} = 2.303 \times nRT \log_{10} \frac{V_2}{V_1} = 2.303 \times nRT \log_{10} \frac{P_1}{P_2} \quad (P_1 V_1 = P_2 V_2)$$

$$W \Rightarrow P_1 V_1 \log_e \frac{V_2}{V_1} = P_2 V_2 \log_e \frac{V_2}{V_1} \quad [P_1 V_1 = P_2 V_2 = nRT]$$

$$\Rightarrow P_1 V_1 \log_e \frac{P_1}{P_2} = P_2 V_2 \log_e \frac{P_1}{P_2}$$

all in above are also equal to Q ($Q = W$ as $\Delta U = 0$)

② Adiabatic process

- constant heat energy) no exchange, $\Delta Q = 0$ ideal gas: $PV^\gamma = \text{constant}$
- Perfectly adiabatic, very fast $P_1 V_1^\gamma = P_2 V_2^\gamma$ $\gamma = \frac{C_P}{C_V}$

Ex: Tyre if burst $\rightarrow$ cools

- As bursting is quick process
- Upon bursting work done = by the gas $\rightarrow$ +ve
- $-\Delta U = W \rightarrow \Delta U \downarrow$ or $\Delta T \downarrow$

$$Q = W + \Delta U$$

or, $-\Delta U = W$

- Perfectly insulated wall

For adiabatic process: $PV^\gamma = \text{constant}$

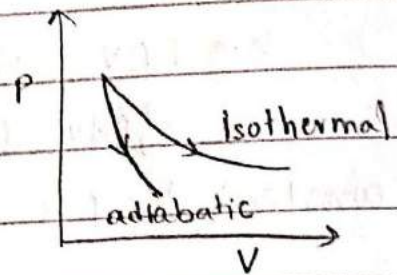
so, $\frac{nRT}{V} V^\gamma = \text{constant}$

$TV^{\gamma-1} = \text{constant}$

$$PV^\gamma = \text{constant}$$

$$P \left(\frac{nRT}{P} \right)^\gamma = \text{constant}$$

$$P^{1-\gamma} T^\gamma = \text{constant}$$



- Slope of adiabatic curve, steeper than isothermal graph

$$(PV) = k \rightarrow \text{isothermal}$$

$$PV^\gamma = k \rightarrow \text{adiabatic}$$

$$d(PV^\gamma) = 0$$

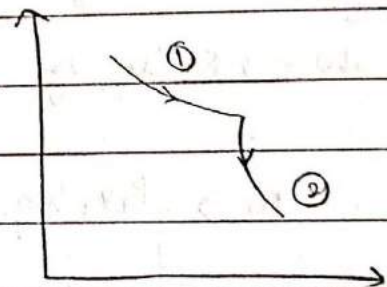
$$P dV^\gamma + V^\gamma dP = 0$$

$$P \gamma V^{\gamma-1} dV + V^\gamma dP = 0$$

$$V^\gamma dP = -P \gamma V^{\gamma-1} dV$$

$$V^\gamma \frac{dP}{dV} = -\frac{P \gamma V^{\gamma-1} dV}{V}$$

$$\left(\frac{dP}{dV} \right)_{\text{adi}} = -\gamma \left(\frac{P}{V} \right)$$



① - isothermal

② - Adiabatic

$$\text{or, } \left(\frac{dP}{dV} \right)_{\text{adiabatic}} = \gamma \left(\frac{dP}{dV} \right)_{\text{iso}} \quad [\text{slope adi} = \gamma \times \text{slope iso}]$$

$$\gamma = 1 + \frac{2}{f}$$

$$\gamma_m = 1.67$$

$$\gamma_d = 1.4$$

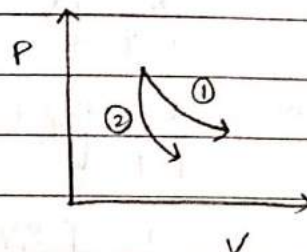
$$\gamma_p = 1.33$$

$$\gamma_m > \gamma_d > \gamma_p$$

$\therefore$ slope of mono > di > poly for adiabatic process - $\frac{PV^\gamma}{V}$

Q) JEE 2002 If ① and ② are P-V slopes of an adiabatic process, such that $\text{slope}_2 > \text{slope}_1$. What are the gases?

- a) He O₂
- b) O₂ He
- c) He Ar
- d) O₂ N₂



slope of adiabatic process $\frac{\Delta P}{\Delta V}$ depends on γ directly,
 $\gamma_m > \gamma_{di}$

$\therefore \text{slope}_m > \text{slope}_{di} \quad \therefore \text{gas 2} = \text{mono}, \text{ gas 1} = \text{di}$

Q) JEE 2010 Diatomic ideal gas is compressed adiabatically to $\frac{1}{32}$ of its initial volume. If initial temp of gas is T_i & final temp is aT_i , the value of a is :-

- a) 1
- b) 2
- c) 3
- d) 4

$$PV^\gamma = \text{constant} \Rightarrow \frac{nRT}{V} V^\gamma = \text{const} \Rightarrow TV^{\gamma-1} = \text{constant}$$

$$V_1 \rightarrow V_2 = \left(\frac{V_1}{32}\right)$$

$$T_1 V_1^{\gamma-1} = aT_i \left(\frac{V_1}{32}\right)^{\gamma-1}$$

$$\cancel{V_1^{\gamma-1}} = a \cancel{V_1^{\gamma-1}} \Rightarrow 32^{\gamma-1} = a = \frac{11^{\frac{2}{5}}}{32} = \frac{2^{1/5}}{32} = 2^2 = 4$$

Work done in adiabatic process

$$P_1, V_1, T_1 \rightarrow P_2, V_2, T_2$$

$$W = \int P dV$$

$$= \int \frac{nRT}{V} dV \rightarrow \text{not useful as } T \text{ not constant}$$

but, $PV^\gamma = k$ or, $P = \frac{k}{V^\gamma}$

$$W = \int_{V_1}^{V_2} \frac{k}{V^\gamma} dV = k \int_{V_1}^{V_2} \frac{1}{V^\gamma} dV = k \int_{V_1}^{V_2} V^{-\gamma} dV$$

$$= k \left(\frac{V^{-\gamma+1}}{-\gamma+1} \right)_{V_1}^{V_2} \rightarrow \text{PRO}$$

$$W = K \left[\frac{V^{-\gamma+1}}{-\gamma+1} \right]_{V_1}^{V_2}$$

$$= \frac{K}{1-\gamma} \left[V_2^{1-\gamma} - V_1^{1-\gamma} \right]$$

$$W = \frac{K}{1-\gamma} \left[V_2^{1-\gamma} - V_1^{1-\gamma} \right] \quad K = P_1 V_1^\gamma = P_2 V_2^\gamma$$

$$= \frac{1}{1-\gamma} \left[\frac{P_2 V_2^\gamma \cdot V_2^{1-\gamma}}{\gamma} - \frac{P_1 V_1^\gamma \cdot V_1^{1-\gamma}}{\gamma} \right] = \frac{1}{1-\gamma} [P_2 V_2 - P_1 V_1]$$

$$W = \frac{P_2 V_2 - P_1 V_1}{1-\gamma} = \frac{nR(T_2 - T_1)}{1-\gamma}$$

$$\text{Or, } -\Delta U = W \therefore \Delta U = -W$$

$$\Delta U = \frac{P_2 V_2 - P_1 V_1}{\gamma - 1} = \frac{nR(T_2 - T_1)}{\gamma - 1}$$

Q) 2011 5.6L of He at STP adiabatically compressed to 0.7L. Taking initial temp to be T_1 , work done in process is

- a) $\frac{9}{8} RT_1$ b) $\frac{3}{2} RT_1$ c) $\frac{15}{8} RT_1$ d) $\frac{9}{2} RT_1$

$$\text{No. of moles } n = \frac{5.6}{22.4} = \frac{1}{4} \text{ moles}$$

$$V_1 = 5.6L \quad V_2 = 0.7L \quad T_1$$

$$W = \int P dV = \int \frac{K}{V^\gamma} dV$$

$$W = \frac{K}{1-\gamma} \left[\frac{V^{-\gamma+1}}{-\gamma+1} \right]_{V_1}^{V_2}$$

$$= \frac{K}{1-\gamma} \left[0.7^{1-\gamma} - 5.6^{1-\gamma} \right]$$

$$W = \frac{nR(T_2 - T_1)}{\gamma - 1}$$

$$TV^{\gamma-1} = \text{constant}$$

$$T_1 \rightarrow V_1 (5.6)$$

$$T_2 \rightarrow V_2 (0.7) = \frac{1}{8} V_1 = \frac{V_1}{8}$$

$$T_1 V_1^{\gamma-1} = T_2 \left(\frac{V_1}{8} \right)^{\gamma-1}$$

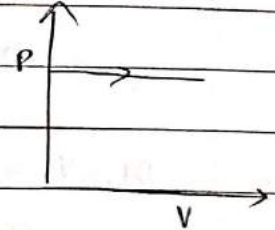
$$\Rightarrow 4T_1 = T_2$$

$$W = \frac{nR(T_2 - T_1)}{\gamma - 1} = \frac{0.25 \lambda R (4T_1 - T_1)}{\frac{2}{3} - 1} = \frac{1}{4} \times \frac{R}{2} \times 3T_1 \times 3 = \frac{9}{8} RT_1$$

③ Isobaric Process - ($\Delta P = 0$)

→ constant pressure - P

$$W = \int_{V_1}^{V_2} P dV = P \int_{V_1}^{V_2} dV = P(V_2 - V_1) = P \Delta V$$



$$W = P \Delta V$$

$$Q = \Delta U + P \Delta V$$

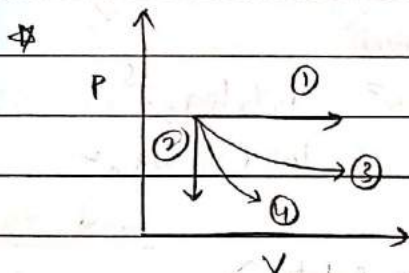
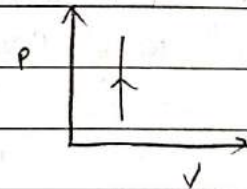
④ Isochoric Process - ($\Delta V = 0$)

$$W = \int P dV = 0$$

$$Q = W + \Delta U$$

$$Q = \Delta U$$

all heat used to ↑ Internal E.



① - Isobaric

② - Isochoric

③ - Isothermal

④ - Adiabatic

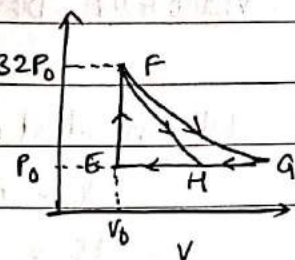
JEE Adv 2017 Q)

1:28:11

JEE Adv 2013) one mole of monoatomic ideal gas, is

moved from $G \rightarrow E$, $G \rightarrow H$, $H \rightarrow F$, $F \rightarrow G$,

$E \rightarrow F$



List I (Process)

List II (Work done)

p) $G \rightarrow E$

q) $G \rightarrow H$

r) $F \rightarrow H$

s) $F \rightarrow G$

$16 P_0 V_0 \log_e 2$

$36 P_0 V_0$

$24 P_0 V_0$

$31 P_0 V_0$

1) GF is isothermal curve

work in GE is $P \Delta V$, but V_2 is not given, from GF $P_1 V_1 = P_2 V_2$

$$32 P_0 V_0 = P_0 (32 V_0)$$

$$\therefore \text{Work} = P_0 \times (V_2 - V_1) = 31 P_0 V_0$$

2) $G \rightarrow H$ is also isobaric

$\therefore W = P \Delta V$ but $V_2 =$ not given

we found $V_1 = 32V_0$

$\rightarrow$ as FH is adiabatic curve $PV^\gamma = \text{constant}$ $\gamma = \frac{5}{3}$

$$32P_0 V_0^{5/3} = P_0 (xV_0)^{5/3}$$

$$\text{or, } 32 = x^{5/3}$$

$$32^{3/5} = x = 8$$

$$\text{or, } V_2 = 8V_0$$

$$\therefore \text{work} = P \Delta V = P_0(32V_0 - 8V_0) = P_0 \times 24V_0 = 24P_0V_0$$

3) $F \rightarrow H$ is adiabatic

$$W = \frac{P_2 V_2 - P_1 V_1}{\gamma - 1} = \frac{(P_0 \times 8V_0) - (32P_0 V_0)}{\frac{2}{5}} = \frac{24P_0 V_0 \times 5}{2} = 60P_0 V_0$$

4) $F \rightarrow G$ is isothermal

$$W = nRT \log_e \frac{V_2}{V_1} = P_1 V_1 \log_e \frac{V_2}{V_1} = 32P_0 V_0 \log_e \frac{32V_0}{8V_0}$$

$$= 32P_0 V_0 \log_e 2^5$$

$$= 160P_0 V_0 \log_e 2$$

Video #07 Derivation of $C_p - C_v = R$ (Mayer's Relation)

case 1 - Let an ideal gas be heated at constant 'V'

$$Q = nC_v \Delta T$$

$$dQ = nC_v dT$$

$$= 1 \times C_v dT$$

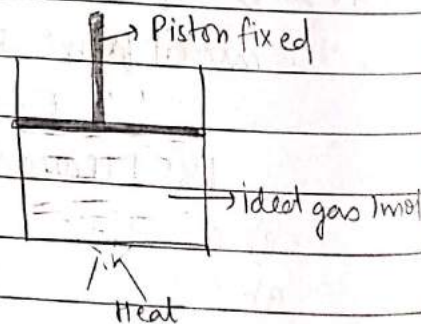
$$W = \int P dV = 0 \quad (\Delta V = 0)$$

$$Q = W + \Delta U$$

$$dQ = dW + dU$$

$$C_v dT = 0 + dU$$

$$dU = C_v dT$$



Case 2 - the pressure is kept fixed now, volume can change such that the same ΔT occurs.

$$Q = n C_p \Delta T$$

$$= C_p \Delta T$$

$$dw = PdV$$

$$dq = du + dw$$

$$C_p dT = du + PdV$$

$$C_p dT = C_v dT + PdV$$

For ideal gas, ΔU only dependent on ΔT and nothing else $\therefore du = C_v dT$

$$(now, PV = nRT \mid PV = RT)$$

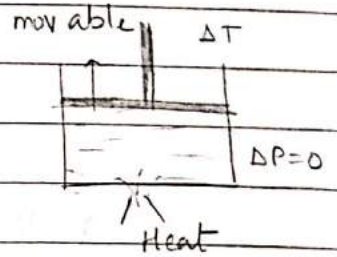
$$PdV + VdP = RT$$

$$\Delta P = 0 \quad PdV = RT$$

$$C_p dT = C_v dT + R dT$$

$$C_p = C_v + R$$

$$\underline{C_p - C_v = R}$$

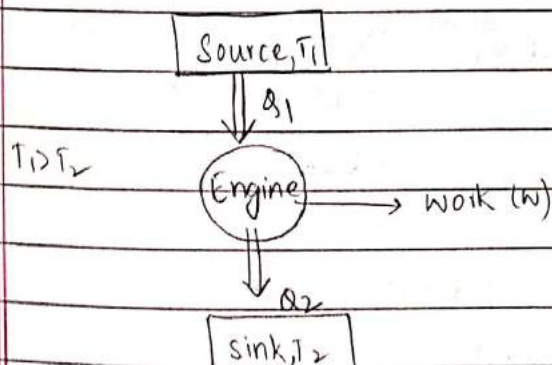


Video #08 - Second law of Thermodynamics
(About spontaneity of a process)

- ① All work can be converted into heat but,
- ② All heat can't be converted into work (used in engine)

Heat Engine -

Makes work from heat (petroleum ignited, to convert into work)



• Source provides some heat Q_1 , out of which some is used to do work w , and other residual energy Q_2 is sent to sink.

• Temp source $T_1 >$ Temp sink T_2

• Efficiency $\eta = \frac{\text{Work done}}{\text{Heat supplied}} \times 100$

$$Q_1 = w + Q_2$$

$$Q_1 - Q_2 = w$$

$$= \frac{w}{Q_1} \times 100$$

$$\text{or, } \frac{Q_1 - Q_2}{Q_1} \times 100$$

$$\text{or, efficiency } \eta = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1}$$

$$\eta = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1}$$

$$\eta\% = \frac{W}{Q_1} \times 100 = \frac{Q_1 - Q_2}{Q_1} \times 100 = \left(1 - \frac{Q_2}{Q_1}\right) \times 100$$

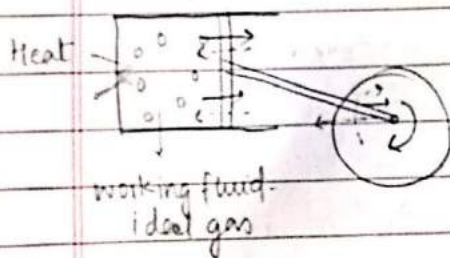
→ $\eta = 100\%$ or, 1 only if $Q_2 = 0$, i.e. heat can't be fully extracted for work

→ For $Q_2 = 0$ the source can be at any T and sink ^{will} be not needed

But not possible

→ There has to be little wastage of heat ($Q_2 \neq 0$)

Working of Engine -

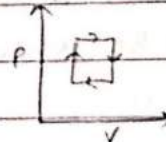
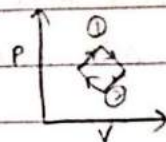
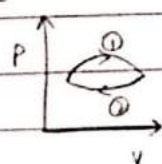


→ When the gas expands, piston moves out and work done to move front

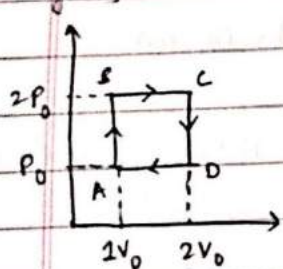
→ Then due to inertia, rod moved back and gas compresses

→ one cycle completed (1 exp + 1 comp)

graphs -



[Heat Engine]



A → B
B → C
C → D
D → A

ΔU
 $\frac{3}{2}RT_0$
 $\frac{3}{2}R(2T_0)$
 $\frac{3}{2}R(-2T_0)$
 $\frac{3}{2}R(-T_0)$

W
0
 $2P_0V_0$
0
 $P_0(V_0) = -RT_0$

Q
 $\frac{3}{2}RT_0$
 $\frac{5}{2}RT_0$
 $-3RT_0$
 $-2.5RT_0$

Heat gain by engine to work (W)
Heat loss after work (W)

1 mole monoatomic gas

$$A \rightarrow B \quad \Delta U = nC_v \Delta T = n \left(\frac{5}{2}R\right) \Delta T = \frac{3}{2}R \Delta T$$

$$PV = nRT \text{ always} \quad \therefore P_0 V_0 = nRT$$

$$\text{then } 2P_0 \cdot 2V_0 = nRT \quad \text{or } T_2 = 2T_0$$

$$\frac{nRT_1}{P_0} = \frac{nRT_2}{2P_0} \quad (V_0 \text{ same})$$

$$2T_1 = T_2$$

$$\text{At A - } P_0 V_0 T_0 \quad \text{B - } P_0 V_0 2T_0 \quad \text{C - } 2P_0 2V_0 4T_0 \quad \text{D - } P_0 2V_0 2T_0$$

$$\text{Net } \Delta U \text{ in 1 cycle} = 0$$

$$\text{work done} = RT_0$$

forward exp -
return comp

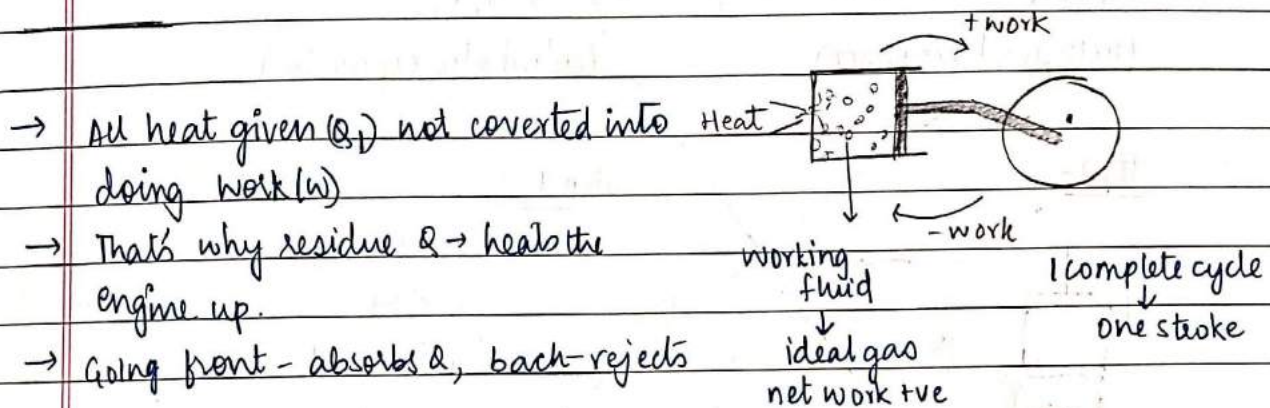
$$Q \text{ lost total} = 5.5RT_0$$

as work = RT_0 and total = $6.5RT_0$

Thermodynamics - 09

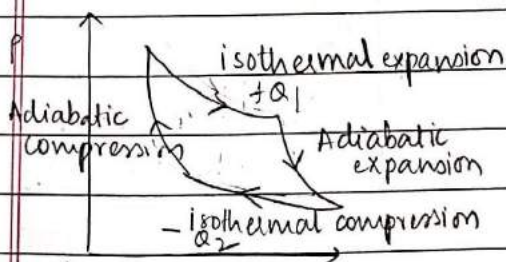
Carnot's Engine

- Carnot's Heat Engine
- Working, graph
- Derivation of $\eta = 1 - \frac{T_2}{T_1}$ with depth
- Refrigerator
- Numericals



Carnot's Engine - Hypothetical engine

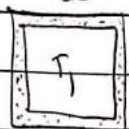
Most effective engine $\rightarrow$ Reversible Engine $\rightarrow \eta = \text{max of any such engine which works b/w same two temp } T_1 \text{ \& } T_2$



$$\eta = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1}$$

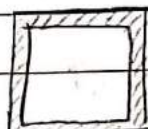
STROKE OF CARNOT'S ENGINE

$\Delta T = 0$ or
Heat capacity
 ∞



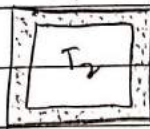
Source

perfectly
conducting



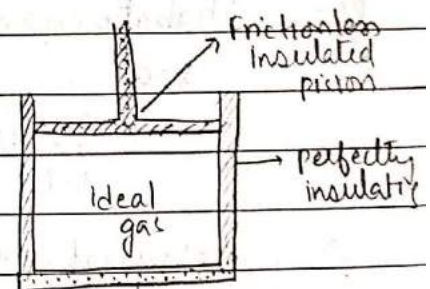
stand

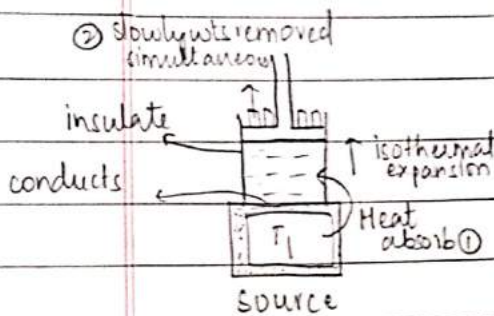
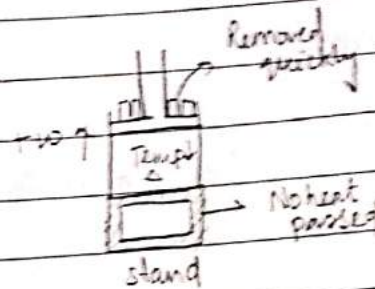
perfectly
insulating



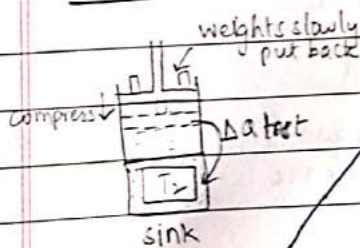
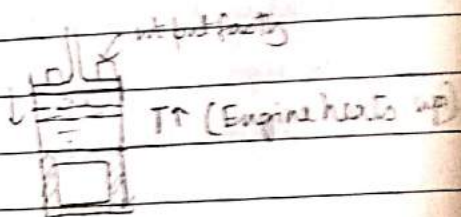
Sink

perfectly
conducting



Step 1-Step 2-

$P_1 V_1 T_1 \xrightarrow{+ve \text{ work}} P_2 V_2 T_2$
(isothermal expansion) (adiabatic expansion)

Step 3-Step 4-

$P_3 V_3 T_1 \xrightarrow{-ve \text{ work}} P_4 V_4 T_2$
(isothermal compression) (adiabatic compression)

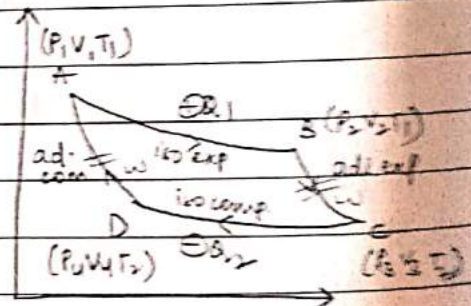
AB- Isothermal expansion

$$\Delta T = 0 \quad \Delta U = 0$$

$$\Delta Q = W + \Delta U \Rightarrow 0$$

$$\Delta Q = W$$

$$Q_1 = W_1 = nRT_1 \log_e \frac{V_2}{V_1}$$

BC- Adiabatic expansion

$$Q = 0$$

$$W_2 = \frac{nR(T_2 - T_1)}{1 - \gamma} = -\Delta U$$

DA- Adiabatic comp

$$Q = 0$$

$$W_4 = \frac{nR(T_1 - T_2)}{1 - \gamma} = -\Delta U$$

CD- Isothermal compression

$$\Delta U = 0$$

$$Q_2 = W_3 = nRT_2 \log_e \frac{V_4}{V_3}$$

$$\text{Total } \Delta U = 0 \quad (W_2 = -W_4)$$

$$\text{Total } W = W_1 + W_2 + W_3 + W_4 = W_{\text{net}}$$

$$\frac{Q_2}{Q_1} = \frac{T_2}{T_1}$$

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$$\eta = \text{Efficiency} = \frac{W_{\text{net}}}{Q_1}$$

$$\eta = \frac{nRT_1 \log_e V_2/V_1 + nRT_2 \log_e V_4/V_3}{nRT_1 \log_e V_2/V_1}$$

$$= 1 + \frac{\log_e V_4/V_3 \times T_2}{\log_e V_2/V_1 \times T_1}$$

$$\text{now, } P_1 V_1 = P_2 V_2 \quad \text{--- (1)}$$

$$P_2 V_2^\gamma = P_3 V_3^\gamma \quad \text{--- (2)}$$

$$P_3 V_3 = P_4 V_4 \quad \text{--- (3)}$$

$$P_4 V_4^\gamma = P_1 V_1^\gamma \quad \text{--- (4)}$$

multiply (1) × (2) × (3) × (4)

$$\frac{P_1 P_2 P_3 P_4}{V_1^{\gamma-1} V_2^{\gamma-1} V_3^{\gamma-1} V_4^{\gamma-1}} \times V_1^\gamma V_2^\gamma V_3^\gamma V_4^\gamma = \frac{P_1 P_2 P_3 P_4}{V_1^{\gamma-1} V_2^{\gamma-1} V_3^{\gamma-1} V_4^{\gamma-1}} \times V_1^\gamma V_2^\gamma V_3^\gamma V_4^\gamma$$

$$V_2^{\gamma-1} V_4^{\gamma-1} = V_3^{\gamma-1} V_1^{\gamma-1}$$

$$V_2 V_4 = V_3 V_1$$

$$\frac{V_2}{V_1} = \frac{V_3}{V_4} \Rightarrow \eta = 1 + \frac{T_2}{T_1} \times \frac{\log_e \frac{V_4}{V_3}}{\log_e \frac{V_2}{V_1}}$$

$$\left[\text{but, } \frac{V_2}{V_1} = \frac{V_3}{V_4} \text{ or, } \frac{V_2}{V_1} = \left(\frac{V_4}{V_3} \right)^{-1} \right]$$

$$\Rightarrow \eta = 1 + \frac{T_2}{T_1} \times \frac{\log_e \left(\frac{V_3}{V_4} \right)^{-1}}{\log_e \left(\frac{V_2}{V_1} \right)}$$

$$= 1 - \frac{T_2}{T_1} \frac{\log_e V_4/V_3}{\log_e V_1/V_2} \Rightarrow \boxed{1 - \frac{T_2}{T_1} = \eta}$$

1Q) Source has $T = 227^\circ\text{C}$, sink $= 127^\circ\text{C}$, Heat absorbed from source $= 10^4 \text{ J}$

work done per cycle

a) 1000 J

b) 2000 J

c) 3000 J

d) 4000 J

$$\eta \Rightarrow 1 - \frac{T_2}{T_1} = \frac{W}{Q_1}$$

$$\Rightarrow 1 - \frac{500}{500} = \frac{W}{10^4} \Rightarrow \frac{10^4}{5} = W \Rightarrow 0.2 \times 10^4 \Rightarrow 2000 \text{ J}$$

Q2) The $\eta = \frac{1}{6}$. If sink Temperature is reduced by 65°C , $\eta = \frac{1}{3}$. Calculate actual sink temperature a) 48°C b) 49°C c) 51°C d) 52°C

$$\eta = \frac{1}{6} = 1 - \frac{T_2}{T_1}$$

$$\frac{T_2}{T_1} = \frac{5}{6}$$

$$\eta' = \frac{1}{3} = 1 - \frac{(T_2 - 65)}{T_1}$$

$$\frac{2}{3} = \frac{T_2 - 65}{T_1} \quad \frac{2T_1 - 130}{2T_1} = \frac{4}{6}$$

$$2T_1 - 130 = T_2$$

$$T_2 = 130$$

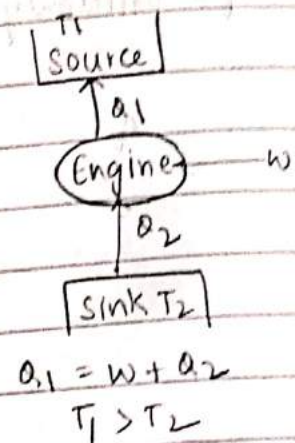
$$\text{or, } T_1 = \frac{T_2 \times 6}{5} = \frac{130 \times 6}{5} = 156$$

solve, $T_2 = 52$

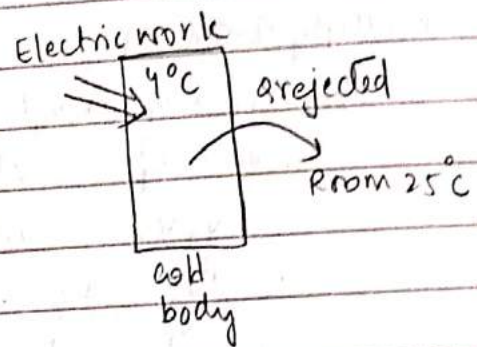
REFRIGERATOR

"Reverse Heat Engine"

- It uses electrical energy (to do work) and extracts heat from cold body (fridge = sink)
- Moves this heat to engine which uses energy and Removes total energy to surroundings ($Q_2 > Q_1$) → Room/Source



- Removing heat from cold to hot body is a non-spontaneous process, Hence needs excess electrical work
- Every thing else is same w/cannot



$$\frac{Q_2}{Q_1} > \frac{T_2}{T_1}$$

Q) If inside refrigerator $T = -3^\circ\text{C}$, Room Temp $= 27^\circ\text{C}$, How much heat is rejected to room per second if 1J of Electrical Energy is given to refrigerator per second?

- a) 4J b) 7J c) 9J d) 13J

Heat rejected = Q_1

$$\frac{T_2}{T_1} = \frac{Q_2}{Q_1} \Rightarrow \frac{Q_1}{Q_2} = \frac{T_1}{T_2} \times Q_2 = \frac{300}{270} \times 1 = \frac{10}{9}$$

$$Q_1 - Q_2 = w = 10 - 9 = 1\text{J} \quad \therefore \frac{w}{Q_1} = \frac{10-9}{9} = \frac{10-9}{9} = \frac{1}{9} \quad \boxed{Q_1 = 9}$$

Coefficient of Performance

Heat Rejected should be ↑,
Work done ↓

$$K = \frac{Q_2}{w} = \frac{Q_2}{Q_1 - Q_2}$$

$$K = \frac{T_2}{T_1 - T_2}$$

For work done in a Polytropic Process

Any process in which $PV^\alpha = \text{constant}$ ($\alpha = \text{anything}$)

→ $\alpha = \gamma$, $PV^\gamma = \text{constant}$ or adiabatic

→ $\alpha = 1$, $PV = \text{constant}$ or isothermal

→ $\alpha = 0$, $PV^0 = P = \text{constant}$ or isobaric

Now work done $\Rightarrow$

$$W = \int P dV$$

$$P = \frac{K}{V^\alpha}$$

$$W = \int_{V_1}^{V_2} \frac{K}{V^\alpha} dV$$

$$= K \int_{V_1}^{V_2} V^{-\alpha} dV = K \left[\frac{V^{-\alpha+1}}{-\alpha+1} \right]_{V_1}^{V_2} = K \left(\frac{V_2^{1-\alpha}}{1-\alpha} - \frac{V_1^{1-\alpha}}{1-\alpha} \right)$$

$$W = K \left(\frac{V_2^{1-\alpha} - V_1^{1-\alpha}}{1-\alpha} \right) \quad K = P_1 V_1^\alpha = P_2 V_2^\alpha$$

$$W = \frac{nR(T_2 - T_1)}{1-\alpha} = \frac{P_2 V_2 - P_1 V_1}{1-\alpha} \quad \therefore W = \frac{nR\Delta T}{1-\alpha}$$

Heat capacity

$$\Delta Q = W + \Delta U$$

$$nC_{\text{poly}} \Delta T = \frac{nR\Delta T}{(1-\alpha)} + nC_V \Delta T$$

$$\frac{C_P}{C_V} = \gamma \Rightarrow C_P = \gamma \cdot C_V$$

$$C_{\text{poly}} = \frac{R}{1-\alpha} + C_V$$

$$\therefore C_{\text{poly}} = \frac{R}{1-\alpha} + C_V$$

$$C_V = \left(\frac{f}{2} R \right)$$